Explicit L^∞ stability bounds for undecimated Daubechies smoothing and wavelet-filtered persistence landscapes

Lean-checked bound propagation and an exact level-one `db2` operator norm

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In topological data analysis, signals may be smoothed by an undecimated wavelet operator before persistent homology and persistence landscapes are computed. We study the composed map obtained by first applying an undecimated low-pass approximation operator and then restricting an indexed persistence landscape to a bounded parameter interval U. Under the standard bottleneck-stability and landscape non-expansiveness assumptions, we obtain, for every landscape index $n \geq 1$ and every homological degree,

$$\|\lambda_n^U(P_j^\Psi Z_1) - \lambda_n^U(P_j^\Psi Z_2)\|_{L^2(U)} \leq \|P_j^\Psi\|_{L^\infty \to L^\infty} \, \|Z_1 - Z_2\|_{L^\infty(D)} \, \sqrt{|U|}.$$

Thus, the only implementation-dependent factor is the L^∞ operator norm of the smoothing map. On a finite grid this norm is the maximum absolute row sum of the corresponding matrix. For the separable, level-one undecimated `db2` operator on a periodic grid of side $N \geq 7$, the one-dimensional approximation kernel is $\frac{1}{32}[-1, 0, 9, 16, 9, 0, -1]$, and the exact finite-grid L^∞ operator norm in dimension d is $(9/8)^d$. We verify this level-one calculation and the algebraic propagation of the abstract stability hypotheses in Lean 4.33.0 against mathlib v4.33.0. The Lean development treats bottleneck stability and landscape non-expansiveness as explicit hypotheses rather than formalizing their underlying topological proofs.

We also report reproducible numerical checks on synthetic fields and an application illustration using 40 public chest radiographs with artificially generated bounded pixel

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perturbations. In the radiograph experiment, all 200 weighted-discrete ratios were below one; the median was 0.149 and the maximum was 0.382. These experiments test the implementation and illustrate the slack in the theoretical bound, but they are not substitutes for the theorem and do not establish a statistical or clinical robustness claim.

Keywords: persistence landscapes; bottleneck stability; wavelet frames; stationary wavelet transform; cubical persistence.

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1 Introduction

1.1 Motivation

Bottleneck stability [11], interleaving stability of persistence modules [6, 23], and non-expansiveness of persistence landscapes [2] provide the standard ingredients [9] for controlling

$$Z \longmapsto \mathrm{Dgm}(Z) \longmapsto \lambda_n(Z).$$

When a smoothing operator P_j^Ψ is inserted before persistence and the resulting landscape is restricted to a bounded interval U, these existing ingredients can be composed with an L^∞ operator bound for P_j^Ψ and the elementary embedding $L^\infty(U) \hookrightarrow L^2(U)$. The resulting estimate is mathematically straightforward, but its numerical usefulness depends on stating the implementation-dependent operator norm explicitly rather than hiding it in an unspecified Lipschitz constant.

This paper carries out that composition for a specified undecimated `db2` low-pass implementation. Its main concrete contribution is the exact level-one L^∞ operator norm on a finite grid, together with reproducible per-scale row-sum computations and a Lean-checked verification that the stated constants propagate correctly through the abstract inequalities.

We prove an $L^2(U)$ bound for this composed map, with $C_{\mathrm{fp}}(j) = \|A_{j+1}\|_{L^\infty \to L^\infty}$ and $\sqrt{\mathrm{diam}(U)}$ left explicit. On a finite periodic grid the constant can be saturated under the explicit conditions of proposition 3.3.

In modern topological machine learning, it is standard practice to smooth signals via stationary wavelet transforms before computing topological summaries [28, 19]. The result below should therefore be read as an explicit and implementation-aware specialization of classical stability theory, not as a new bottleneck- or landscape-stability theorem. Its practical value is that every factor in the resulting certificate is stated and can be evaluated for the implemented finite-grid operator.

1.2 Constants in the composed bound

Two factors appear once the classical $L^\infty(\mathbb{R})$ bound is rewritten for $\lambda_n|_U \circ P_j^\Psi$. The smoothing operator P_j^Ψ is not an isometry: it multiplies the L^∞ norm by $C_{\mathrm{fp}}(j) = \|P_j^\Psi\|_{L^\infty \to L^\infty}$. Restricting

to U converts $L^\infty(\mathbb{R})$ control into $L^2(U)$ control through $\|g\|_{L^2(U)} \le \sqrt{\operatorname{diam}(U)}\,\|g\|_{L^\infty(U)}$. These are the constants in Theorem A.

1.3 Results

1. Theorem A: stability of $Z \mapsto \lambda_n|_U \circ P_j^\Psi Z$ in $L^2(U)$, with $C_{\mathrm{fp}}(j)$ and $\sqrt{\operatorname{diam}(U)}$ explicit, for every $n \ge 1$ and every homological degree, saturating the constant under the explicit conditions of proposition 3.3.
2. Proposition 3.4: for the implemented separable D4 (`db2`) operator, $\|P_{0,N}^\Psi\|_{L^\infty \to L^\infty} = (9/8)^d$ on periodic grids of side $N \ge 7$.
3. Numerical checks of the implemented operator and of Theorem A, including a deterministic pair whose computed ratio approaches one to the stated floating-point tolerance (section 6).
4. A medical-imaging illustration (section 6.2): 200 artificial bounded-perturbation comparisons on 40 public chest radiographs produce numerical ratios consistent with the certificate of Theorem A, with the factorization of eq. (11).

Section 2 fixes notation. Section 3 contains Theorem A and the operator-norm propositions. Corollaries are in section 4; the machine-checked formalization in section 5; numerical checks in section 6, including the real-data illustration of section 6.2. An appendix records why a kernel estimate is not used for $C_{\mathrm{fp}}(j)$.

2 Mathematical preliminaries

2.1 Spatial domain versus parameter interval

Two geometric objects must be kept separate. The *spatial domain* is a compact $D \subset \mathbb{R}^d$ carrying input signals $Z : D \to \mathbb{R}$; the wavelet frame acts here. When bottleneck stability of Cohen–Steiner–Edelsbrunner–Harer is invoked, D is triangulable [11]; the computational model of section 6.1 is a cubical complex [21]. The *parameter interval* is a bounded interval $U \subset \mathbb{R}$ (open, closed, or half-open: $L^2(U)$ and $\operatorname{diam}(U)$ see only Lebesgue measure) on which landscapes are restricted; the landscape $\lambda_n : \mathbb{R} \to \mathbb{R}$ is a function of the filtration parameter t, and $\lambda_n|_U$ is the object we control. The $L^2(U)$ norm is the Lebesgue L^2 norm over the interval U, and $\operatorname{diam}(U)$ is its length. For any $g : \mathbb{R} \to \mathbb{R}$,

$$\|g\|_{L^2(U)} \;\le\; \sqrt{\operatorname{diam}(U)}\,\|g\|_{L^\infty(U)}, \tag{1}$$

the elementary inequality underlying every bound below.

Remark 2.1 (Intervals versus unions). Throughout the paper U is a bounded interval, so $\operatorname{diam}(U)$ coincides with its Lebesgue length $|U|$. If U is a finite union of intervals, the same proofs yield the (weakly) sharper factor $\sqrt{|U|}$ in place of $\sqrt{\operatorname{diam}(U)}$. We retain $\operatorname{diam}(U)$ for notational consistency with the Lean development, which uses `Metric.diam`.

2.2 Persistence diagrams and landscapes

With the domain and interval distinguished, we recall the objects built from them. The sublevel filtration of $Z : D \to \mathbb{R}$ is $\{X_t = Z^{-1}((-\infty, t])\}_{t\in\mathbb{R}}$. For homological dimension $k \in \{0, 1\}$ the persistence diagram $\mathrm{Dgm}_k(Z)$ is a multiset of points (b, d) with $b < d$. Under

assumption 2.5 the module is q-tame; on a finite filtered cubical complex the persistence barcode has finitely many intervals, some of which may be essential. The n-th persistence landscape, the functional summary used throughout, is

$$\lambda_n(t) = n\text{-th largest value of } \{\min(t-b,\, d-t)_+\}_{(b,d)\in \mathrm{Dgm}_k(Z)}, \qquad t \in \mathbb{R}, \tag{2}$$

with $x_+ = \max(x, 0)$ [2]. We write $\lambda_n(Z) = \lambda_n$ and $\lambda_n^U(Z) = \lambda_n|_U$.

Remark 2.2 (Finite and essential intervals). The ordinary persistence-landscape formula in eq. (2) is used here for finite intervals only. This is also the convention used in all numerical experiments, where infinite H_0 intervals are discarded before the landscape is constructed. Accordingly, Theorem A is stated for the finite-bar landscape associated with the persistence diagram after removal of essential intervals. An extension including essential intervals would require an explicit convention for half-infinite landscape functions and a corresponding statement of landscape non-expansiveness for that convention. We do not use or claim such an extension in the present paper. Under the usual extended bottleneck metric, a finite diagram point cannot be matched to a point with infinite death at finite cost. Consequently, any finite-cost matching separates finite and essential intervals, and its restriction to the finite subdiagrams has no larger cost. The bottleneck bound used below therefore passes to the finite-bar diagrams.

Remark 2.3 (Homological degree). We emphasize $k \in \{0, 1\}$ only because those degrees dominate the imaging applications we have in mind. The proof of Theorem A uses bottleneck stability and non-expansiveness of the landscape map ($L_\lambda = 1$, see remark 2.4) and extends verbatim to every $k \geq 0$. Cubical persistent homology of the sublevel filtration [21, 18] is the computational model in section 6.1.

Remark 2.4 (Non-expansiveness). We write L_λ for the Lipschitz constant of the map $Z \mapsto \lambda_n$ from $L^\infty(D)$ to $L^\infty(\mathbb{R})$. By Bubenik's theorem [2] composed with bottleneck stability [11] the landscape map is non-expansive: $L_\lambda = 1$ for every $n \geq 1$ and every $k \geq 0$. We keep the symbol explicit because section 7 discusses analogues for other functional summaries (silhouettes, persistence images), where it need not equal 1.

2.3 Undecimated low-pass approximation operator

The implemented operator is derived from the stationary undecimated `db2` construction; standard background on wavelets and frames is available in [13, 24, 10, 5, 17].

We use j to count undecimated low-pass cascade stages. Let $k_0 = \frac{1}{2}R_h$, where R_h is the discrete autocorrelation of the orthonormal `db2` low-pass filter h. For an integer $s \geq 1$, define the dyadic upsampling operator by

$$(\mathrm{up}_s\, k)(m) := \begin{cases} k(m/s), & s \mid m, \\ 0, & s \nmid m. \end{cases}$$

For $j \geq 0$, define the one-dimensional cumulative approximation kernel by

$$k_j := k_0 * (\mathrm{up}_2\, k_0) * \cdots * (\mathrm{up}_{2^j}\, k_0), \tag{3}$$

where for $j = 0$ the right-hand side consists only of k_0. The d-dimensional kernel is the tensor product $K_j = k_j^{\otimes d}$.

With this convention, the support length is

$$L_j = 6(2^{j+1} - 1) + 1.$$

On a periodic finite grid, define

$$P_j^{\Psi} Z := K_j * Z, \tag{4}$$

with circular convolution and the tensor-product convention in dimension d. Thus, increasing j applies a broader low-pass kernel and produces stronger smoothing in the convention used in the numerical section. We also write $P_j^{\Psi} = A_{j+1}$ as notation for this implemented approximation operator; this is not intended to assert the telescoping identity $A_{j+1} = A_{j_0} + \sum_{r=j_0}^{j} D_r$ under every MRA indexing convention. All theoretical and numerical statements below refer to the operational definition through the implemented low-pass kernel. For the implemented `db2` generator, φ denotes the corresponding scaling function.

Theorem A uses only linearity, L^∞ boundedness, and persistence tameness of the filtered fields.

Assumption 2.5 (Persistence admissibility)**.** For every input considered below, both Z and $P_j^{\Psi} Z$ define sublevel-set persistence modules to which the stated bottleneck stability theorem applies. Concretely, they are either continuous functions on a compact triangulable domain satisfying the required tameness hypotheses [8], or functions on the finite cubical complex used in the numerical implementation. On a finite filtered cubical complex, the persistence barcode has finitely many intervals, some of which may be essential.

2.4 Norms

The $L^\infty(D)$ norm is the supremum on D, not the essential supremum: sublevel persistence is not invariant under modification on a null set. On a finite cubical grid this is the maximum absolute value. The $L^2(U)$ norm is Lebesgue L^2 on the bounded interval U, under the standing square-integrability of landscapes on bounded U (lemma A.1).

3 Main results

3.1 Stability of the composed map

Lemma 3.1 (Localized landscape map)**.** *For continuous Z, Z' on compact triangulable D (or functions on a finite cubical grid) and bounded $U \subset \mathbb{R}$, $\|\lambda_n^U(Z) - \lambda_n^U(Z')\|_{L^2(U)} \leq \|Z - Z'\|_{L^\infty(D)} \sqrt{\operatorname{diam}(U)}$.*

The stability statement below is abstract with respect to the smoothing operator: P_j^{Ψ} may act either on continuous functions or on finite-grid signals, provided that it is bounded in the supremum norm and maps the inputs under consideration to fields satisfying the persistence hypotheses. The `db2` convolution defined in eq. (4) is the finite-periodic-grid instance analyzed explicitly in proposition 3.4.

Lemma 3.2 (Operator bound)**.** *Let P_j^Ψ be a bounded smoothing operator on the signal space. Let $C_{\mathrm{fp}}(j)$ be any certified constant satisfying $\|P_j^\Psi f\|_{L^\infty(D)} \le C_{\mathrm{fp}}(j)\|f\|_{L^\infty(D)}$ for every admissible f. The smallest such constant is the operator norm $\|P_j^\Psi\|_{L^\infty\to L^\infty}$.*

On a finite grid, the exact operator norm is the maximum absolute row sum. For a periodic separable convolution it is the ℓ^1 norm of the periodized kernel. The ℓ^1 norm of the corresponding non-periodized kernel is always a valid upper bound and is exact when periodization causes no sign cancellation.

Theorem A (Localized $L^2(U)$ landscape stability)**.** *Let Z_1, Z_2 satisfy assumption 2.5, $\Delta = Z_1 - Z_2$, and $U \subset \mathbb{R}$ a bounded interval. Then for every $j \ge 0$, every $n \ge 1$ and every homological degree $k \ge 0$,*

$$\|\lambda_n^U(P_j^\Psi Z_1) - \lambda_n^U(P_j^\Psi Z_2)\|_{L^2(U)} \;\le\; C_{\mathrm{fp}}(j)\,\|\Delta\|_{L^\infty(D)}\,\sqrt{\mathrm{diam}(U)}, \tag{5}$$

where $C_{\mathrm{fp}}(j)$ is any certified L^∞ operator bound as in lemma 3.2. The sharpest choice is $C_{\mathrm{fp}}(j) = \|P_j^\Psi\|_{L^\infty\to L^\infty}$. The argument uses only assumption 2.5 and the operator bound in lemma 3.2. The landscapes in this statement are constructed from finite persistence intervals; essential intervals are omitted (remark 2.2). Because U is an interval, $|U| = \mathrm{diam}(U)$; we use $\sqrt{|U|}$ below to emphasize that the L^2 conversion depends on Lebesgue measure.

The inequality is a direct composition of three known or elementary bounds: bottleneck stability, non-expansiveness of the indexed landscape, and $\|g\|_{L^2(U)} \le \sqrt{|U|}\,\|g\|_{L^\infty(U)}$. We do not claim novelty for these individual steps or for their abstract composition. The implementation-specific contribution is the explicit evaluation of $\|P_j^\Psi\|_{L^\infty\to L^\infty}$, including the exact closed-form value at the first `db2` level.

Proof. By bottleneck stability [11] and non-expansiveness of the landscape map [2],

$$\|\lambda_n(P_j^\Psi Z_1) - \lambda_n(P_j^\Psi Z_2)\|_{L^\infty(\mathbb{R})} \le \|P_j^\Psi \Delta\|_{L^\infty(D)}. \tag{6}$$

Restricting to bounded U and using $\|g\|_{L^2(U)} \le \sqrt{\mathrm{diam}(U)}\,\|g\|_{L^\infty(U)}$ (eq. (1)) yields

$$\|\lambda_n^U(P_j^\Psi Z_1) - \lambda_n^U(P_j^\Psi Z_2)\|_{L^2(U)} \le \sqrt{\mathrm{diam}(U)}\,\|\lambda_n(P_j^\Psi Z_1) - \lambda_n(P_j^\Psi Z_2)\|_{L^\infty(\mathbb{R})}. \tag{7}$$

Chaining eqs. (6) and (7) and lemma 3.2 gives the claim. □

Proposition 3.3 (Conditional saturation criterion)**.** *Let A be a linear operator on a finite grid, let x_0 be a row attaining $\|A\|_{L^\infty\to L^\infty}$, and let Δ satisfy $\|\Delta\|_\infty = \varepsilon$ and $(A\Delta)(x_0) = -\varepsilon\,\|A\|_{L^\infty\to L^\infty}$. Assume that AZ_1 has an isolated finite H_0 persistence interval $[b, d]$ born at x_0, and assume that, for $Z_2 = Z_1 + \Delta$:*

(i) the same persistence pair remains isolated and matched to $[b, d]$;
(ii) its birth changes from b to $b - \delta$, where $\delta = \varepsilon\,\|A\|_{L^\infty\to L^\infty}$;
(iii) its death value remains d; and
(iv) U is a nonempty bounded interval, no other landscape tent reaches the first landscape on U, and $U \subset \left(b, \frac{b+d}{2}\right) \cap \left(b - \delta, \frac{b-\delta+d}{2}\right)$.

Then

$$\|\lambda_1^U(AZ_1) - \lambda_1^U(AZ_2)\|_{L^2(U)} = \|A\|_{L^\infty\to L^\infty}\,\|Z_1 - Z_2\|_\infty\,\sqrt{|U|}. \tag{8}$$

Proof. Under assumptions (i)–(iv), the two first landscapes on U are respectively $t - b$ and $t - (b - \delta)$. Their pointwise difference is therefore δ. Hence their $L^2(U)$ distance is $\delta\sqrt{|U|}$, and substitution of $\delta = \varepsilon\|A\|_{L^\infty \to L^\infty}$ gives the result. □

The proposition is a sufficient criterion, not an unconditional existence theorem for every grid size, scale, boundary mode, or homological degree. Section 6.1 reports one deterministic finite-grid construction that numerically exhibits the birth shift, unchanged death value, and landscape behavior required by these conditions. A general constructive existence proof is not claimed.

The constants $C_{\mathrm{fp}}(j)$ and $\sqrt{\mathrm{diam}(U)}$ are those of the composed map $Z \mapsto \lambda_n|_U \circ P_j^\Psi Z$ and do not appear in the global $L^\infty(\mathbb{R})$ theorems.

Proposition 3.4 (Exact level-one finite-grid norm and higher-level kernel bounds)**.** *Let k_j be the finitely supported one-dimensional approximation kernel on $\mathbb{Z}$, and let $k_{j,N}$ denote its periodization on a periodic grid of side N,*

$$k_{j,N}(r) := \sum_{q\in\mathbb{Z}} k_j(r + qN).$$

For the separable d-dimensional periodic convolution operator $P_{j,N}^\Psi$,

$$\|P_{j,N}^\Psi\|_{L^\infty\to L^\infty} = \|k_{j,N}\|_{\ell^1}^d \le \|k_j\|_{\ell^1}^d.$$

The first equality is the exact finite-grid operator norm. Equality with $\|k_j\|_{\ell^1}^d$ holds whenever periodization produces no sign cancellation; in particular, it holds when the grid side exceeds the support span of k_j.

At level $j = 0$, k_0 has seven taps. Hence, for $N \ge 7$,

$$\|P_{0,N}^\Psi\|_{L^\infty\to L^\infty} = \left(\tfrac{9}{8}\right)^d. \tag{9}$$

In dimension $d = 2$ this gives the exact value $81/64 = 1.265625$.

For the non-periodized kernels on $\mathbb{Z}^2$, direct evaluation gives

$$(\|k_0\|_{\ell^1}^2, \ldots, \|k_6\|_{\ell^1}^2) \approx (1.265625,\ 1.364151,\ 1.391375,\ 1.398358,\ 1.400115,\ 1.400555,\ 1.400665). \tag{10}$$

These values are grid-independent upper bounds for the corresponding finite periodic operators. They equal the finite-grid operator norms only when the stated no-cancellation condition is verified. The displayed values are rounded numerical evaluations, not exact symbolic expressions.

Proof. On a finite grid the operator is a matrix, so $\|\cdot\|_{L^\infty\to L^\infty}$ is the maximum absolute row sum and is attained at a sign pattern. Under periodization it is shift-invariant, so the norm coincides with the ℓ^1 norm of the periodized impulse response. For the separable periodic operator the response is the outer product of the one-dimensional kernel with itself, hence the d-dimensional row-sum norm is the d-th power of the one-dimensional ℓ^1 norm.

Periodizing the kernel combines coefficients whose indices agree modulo N. The triangle inequality therefore gives $\|k_{j,N}\|_{\ell^1} \le \|k_j\|_{\ell^1}$. Consequently, the non-periodized kernel norm is always a valid finite-grid operator bound, although it need not be the exact finite-grid norm when wrapped coefficients cancel.

Let $h = \frac{1}{4\sqrt{2}}(1+\sqrt{3},\ 3+\sqrt{3},\ 3-\sqrt{3},\ 1-\sqrt{3})$ be the orthonormal `db2` low-pass filter, and write $R_h(m) = \sum_n h_n h_{n+m}$. Orthonormality gives $R_h(0) = 1$; the quadrature-mirror filter (QMF) identity gives $R_h(\pm 2) = 0$. Expanding the remaining lags, $R_h(\pm 1) = \frac{9}{16}$ and $R_h(\pm 3) = -\frac{1}{16}$. The one-dimensional undecimated approximation kernel at the first level is half the autocorrelation, $\frac{1}{2}R_h = \frac{1}{32}[-1, 0, 9, 16, 9, 0, -1]$, with ℓ^1 norm $36/32 = 9/8$. This seven-tap kernel is half the autocorrelation of the compactly supported Daubechies filter [29]; its symbol is $\frac{1}{2}|P_{\mathrm{DD4}}|^2$, where P_{DD4} is the symbol of the four-point Deslauriers–Dubuc interpolatory filter $[-1, 9, 9, -1]/16$ [14]. It is not that four-point filter. Tensorization yields eq. (9). Cascading the same kernel produces the computed row sums of eq. (10).

The same computation for `db`N identifies the level-one norm with the ℓ^1 norm of $\frac{1}{2}R_h$, the half-autocorrelation of the `db`N low-pass filter: in one dimension, `db3` gives $612/512 = 1.1953125$ (table 1).

The statements above concern periodic convolution. Symmetric and zero-padded implementations are treated as separate finite matrices; their exact norms, when used, are obtained from their maximum absolute row sums. No general equality with the periodic norm is asserted beyond the level-one interior-row observation reported in section 7.1. □

Table 1: Level-one $L^\infty \to L^\infty$ row-sum norms $\|A_1\|_{L^\infty \to L^\infty} = \|k_0\|_{\ell^1}^d$ on $\mathbb{Z}^d$. Displayed entries are rounded to six decimal places. Closed-form or rational values are stated explicitly where supplied: `db2` (D4) $9/8$, $81/64$, $(9/8)^3$; `db3` one-dimensional $612/512$. The table shows only the first undecimated level ($j = 0$). The CSV `experiments/results/csv/daubechies_opnorms.csv` also records $j = 0, \ldots, 7$ and $d = 1, 2, 3$.

	$d = 1$	$d = 2$	$d = 3$
`db1` (Haar)	1.000000	1.000000	1.000000
`db2` / D4	1.125000	1.265625	1.423828
`db3`	1.195313	1.428772	1.707829
`db4`	1.244141	1.547886	1.925788

Remark 3.5 (Relation to existing stability results). The present estimate does not improve the bottleneck or landscape Lipschitz constants. Rather, it specializes those results to a composed finite-grid pipeline by supplying an explicit norm for the preceding smoothing operator and retaining the dependence on the measure of U. Multi-scale landscape summaries (persistence flamelets) and kernels on the diagram [27, 28, 19] likewise do not substitute a frame operator on the field. The distinction in table 2 is therefore one of theorem scope and implementation specificity, not logical incompatibility with prior work.

Table 2: Components stated explicitly in the cited results. A dash means that the component is not part of the cited theorem as stated; it does not mean that the result cannot be composed with an independently established operator bound or restriction inequality. ✓ = stated explicitly, — = not stated.

	metric	P_j^Ψ	restrict to U	smoothing-operator constant
Cohen–Steiner et al. [11]	bottleneck	—	—	—
Chazal et al. [6] / Lesnick [23]	interleaving	—	—	—
Bubenik [2]	$L^\infty(\mathbb{R})$	—	—	—
Skraba–Turner [30]	p-Wass.	—	—	—
Bubenik–Zhao [3]	rank-Wass.	—	—	—
This work	$L^2(U)$	✓	✓	✓

4 Corollaries

Removing the localization to U recovers a global $L^\infty(\mathbb{R})$ bound. Squaring the landscape norm yields a bound on the difference of the squared $L^2(U)$ norms.

Corollary 4.1 (Global stability recovered)**.** *Under the hypotheses of Theorem A,*

$$\|\lambda_n(P_j^\Psi Z_1) - \lambda_n(P_j^\Psi Z_2)\|_{L^\infty(\mathbb{R})} \le C_{\mathrm{fp}}(j)\,\|\Delta\|_{L^\infty(D)}.$$

Proof. This is eq. (6) composed with lemma 3.2, and does not pass through the $L^2(U)$ statement: the factor $\sqrt{\mathrm{diam}(U)}$ diverges as U enlarges and cannot be "absorbed". □

Corollary 4.2 (Squared-norm stability)**.** *Let* $\mathcal{T}_n^U(j;Z) := \|\lambda_n^U(P_j^\Psi Z)\|_{L^2(U)}^2$ *and let* $\Gamma_j^U(\Delta)$ *be the right-hand side of Theorem A. Then*

$$\left|\mathcal{T}_n^U(j;Z_1) - \mathcal{T}_n^U(j;Z_2)\right| \le \Gamma_j^U(\Delta)\left(\|\lambda_n^U(P_j^\Psi Z_1)\|_{L^2(U)} + \|\lambda_n^U(P_j^\Psi Z_2)\|_{L^2(U)}\right).$$

Proof. Write $a = \|\lambda_n^U(P_j^\Psi Z_1)\|_{L^2(U)}$, $b = \|\lambda_n^U(P_j^\Psi Z_2)\|_{L^2(U)}$. Then $|a^2 - b^2| = |a-b|(a+b)$, and the reverse triangle inequality (lemma A.1) gives $|a-b| \le \Gamma_j^U(\Delta)$ by Theorem A. Multiply. □

5 Lean formalization of the algebraic bound propagation

We formalized two components in Lean 4.33.0 against mathlib v4.33.0 [15, 25].

First, an abstract theorem verifies that a supplied L^∞ landscape-Lipschitz hypothesis, a supplied L^∞ operator bound, and the standard $L^\infty(U)$-to-$L^2(U)$ estimate imply the stated composed inequality with the factors $C_{\mathrm{fp}}(j)$ and $\sqrt{\mathrm{diam}(U)}$. Bottleneck stability, the construction of persistence diagrams, and non-expansiveness of persistence landscapes are not proved in this development; they enter as named hypotheses.

Second, the module `TWS/DaubechiesOpNorm.lean` verifies the algebraic identities used for the level-one orthonormal `db2` filter and proves that the corresponding one-dimensional seven-tap

kernel has ℓ^1 norm $9/8$. It also verifies the tensor-product calculation giving the two-dimensional value $(9/8)^2$.

Thus, the formal artifact certifies the stated algebraic composition and the exact level-one filter calculation, conditional on the explicitly declared topological hypotheses. It does not constitute a formalization of the complete persistent-homology pipeline or of the numerical experiments.

The foundational axioms required for these declarations are strictly limited to `propext`, `Classical.choice` and `Quot.sound`. The full source code currently resides in the project source tree under `formalization/TWS/`. It will be made publicly available with the archival deposit described in the Data and code availability statement. Related formal work on persistent homology outside Lean is reported in [20].

6 Numerical illustration

Figure 1 shows the implemented approximation A_{j+1} and the first landscape of a projected one-dimensional signal. Section 6.1 records the operator norms and the tightness checks.

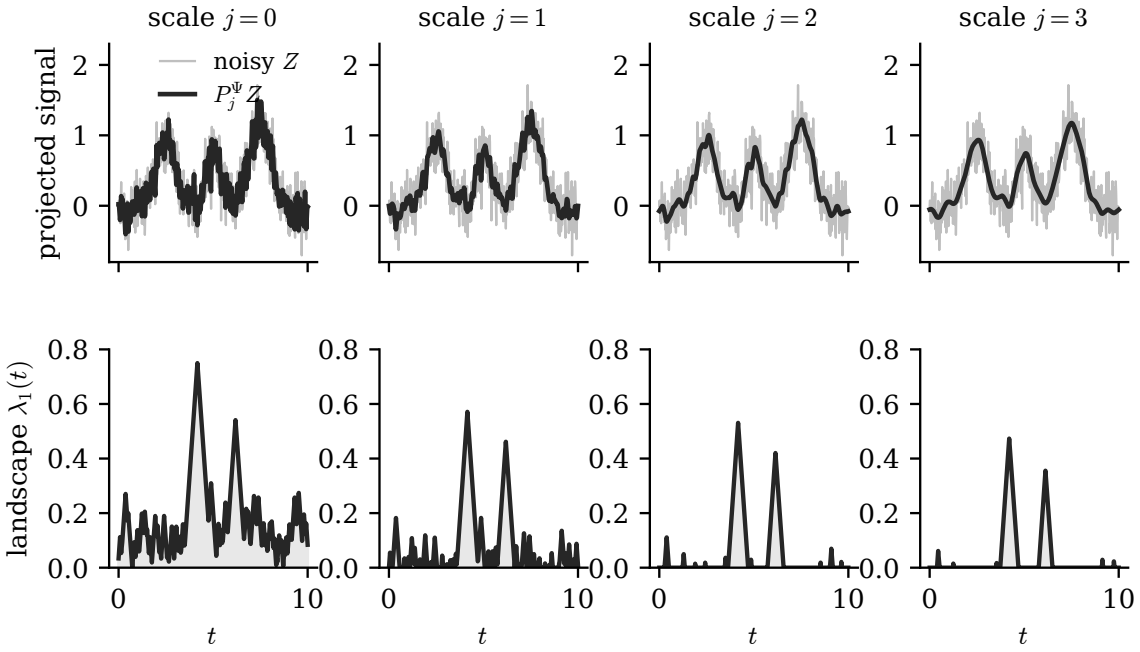


Figure 1: Undecimated D4 (`db2`) approximation A_{j+1} at successive scales. Top: noisy signal and $P_j^{\Psi} Z$. Bottom: corresponding first persistence landscape $\lambda_1(t)$ drawn on the full t-axis (not restricted to a window U). Coarser j suppresses fine oscillations and merges landscape tents.

The constants in section 6.1 belong to one generator: the Daubechies wavelet with two vanishing moments (D4 / PyWavelets `db2`). The checks below use that frame on synthetic fields.

6.1 Reproducible finite-grid checks

The experiments in this section are implementation checks and tightness illustrations. They are not needed to establish Theorem A. Landscape L^2 distances are evaluated numerically on

Table 3: Constants and properties of the implemented two-dimensional `db2` construction. At $j = 0$ and grid side $N \geq 7$, the exact finite-grid operator norm is 81/64. For $j > 0$, the non-periodized-kernel quantities in eq. (10) are certified upper bounds unless an exact finite-grid row sum is reported separately. Theorem A uses only a certified operator bound and the persistence admissibility hypothesis.

	symbol	`db2`
level-one kernel taps	—	7
operator bound (lemma 3.2)	$C_{\mathrm{fp}}(j)$	exactly 81/64 at $j = 0$ for $N \geq 7$; higher-level certified bounds in eq. (10)
landscape Lipschitz	L_λ	1

a finite parameter grid, so the reported ratios are weighted discrete quantities evaluated in floating-point arithmetic.

When landscapes are evaluated on nodes $t_1, \ldots, t_m$, the numerical quantity reported below is the weighted discrete norm

$$\|f\|_{2,w} := \Big(\sum_{i=1}^{m} w_i |f(t_i)|^2\Big)^{1/2},$$

where the nonnegative quadrature weights satisfy $\sum_i w_i = |U|$. This discrete norm obeys the same elementary estimate $\|f\|_{2,w} \leq \sqrt{|U|}\,\|f\|_\infty$. Accordingly, the computed ratios are directly bounded by the same right-hand side, independently of whether the quadrature exactly equals the continuous $L^2(U)$ integral. We nevertheless distinguish the computed discrete norm from the continuous norm appearing in Theorem A.

The theorem and the numerical checks refer to the same generator: D4 / PyWavelets `db2`. Theorem A uses only lemma 3.2 and assumption 2.5.

We implemented the undecimated low-pass approximation operator P_j^Ψ with the undecimated (stationary, translation-invariant) Daubechies wavelet with two vanishing moments (D4 / PyWavelets `db2`), not a Gaussian proxy [12, 26, 4]. Three boundary modes were tested: periodization, symmetric padding, and zero padding.

A ratio maximized over a finite pool of test inputs is a *sample* maximum, so we fix the invocation rather than quote the number alone: 128×128 fields, scales $j = 0, \ldots, 6$ (with the scale convention $j \geq 0$ fixed in section 2), seed 20260715, and 71 inputs (64 random sign patterns, 4 unit deltas, and 3 deterministic textures). The largest ratio observed in this finite input pool is 1.2656 at $j = 0$. Independently of that sample maximum, proposition 3.4 proves that the exact periodic finite-grid norm at this level is 1.265625. If the maximizing row-sign input is included in the pool, the sampled ratio equals that norm up to floating-point rounding.

On a finite grid the implemented operator is a matrix; $\|\cdot\|_{L^\infty \to L^\infty}$ is the maximum absolute row sum and is attained at an explicit sign pattern rather than approached; the $d = 2$ value at $j = 0$ is $(9/8)^2 = 1.265625$ in all three boundary modes on the 128×128 grid, because the seven-tap kernel fits in the interior. Under periodization the map is moreover shift-invariant, so the row sum coincides with the ℓ^1 norm of the impulse response; under the two padded modes it is not shift-invariant and the maximizing input differs, yet at $j = 0$ the norm takes the same value. Emulation of the non-periodic modes at coarser j is discussed in section 7.1. For $j > 0$, the non-periodized kernel bounds reported in eq. (10) increase from 1.364151 at $j = 1$ to

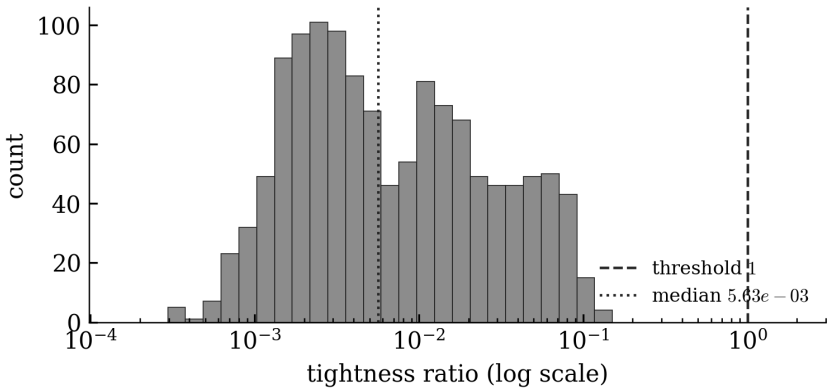


Figure 2: The 1280 values of ρ_w in eq. (11), against the certified per-scale constant $C_{\mathrm{fp}}(j)$ used in each experiment. The dashed line is the threshold 1; the dotted line is the median.

1.400665 at $j = 6$. On a particular finite periodic grid, the exact operator norm is instead the ℓ^1 norm of the periodized kernel and may be smaller because of wrapped-coefficient cancellation. For every experiment below we use either the exact maximum row sum of the implemented finite-grid matrix or the stated non-periodized kernel bound; the accompanying CSV records which constant was used. Random test inputs generally under-reach either quantity because a random sign pattern need not agree with a maximizing matrix row. The reported sample maxima are therefore implementation checks, not operator-norm estimators.

Each experimental CSV row records the columns `grid_shape`, `j`, `boundary`, `constant_type`, and `C_fp_used`. The field `constant_type` takes one of the values `exact_matrix_row_sum`, `exact_periodized_kernel`, or `nonperiodized_kernel_bound`.

With the operator norm measured, we turn to the landscape bound of Theorem A. We ran 1280 landscape comparisons on 48×48 fields with seed 20260715: 160 noise realizations on controlled Gaussian-bump fields, each at four scales $j \in \{0, 1, 2, 3\}$ (again with the same convention $j \geq 0$) and two homological degrees $k \in \{0, 1\}$, using the D4 operator, cubical persistent homology [21, 18], and the weighted discrete landscape norm defined above, applied to the first landscape over $U = (-1, 1)$. Using the stated discretization and floating-point implementation, none of the 1280 computed ratios exceeded one. This observation checks consistency with the theorem but does not provide an independent proof of it. Write

$$\rho_w := \frac{\|\lambda_n^U(P_j^\Psi Z_1) - \lambda_n^U(P_j^\Psi Z_2)\|_{2,w}}{C_{\mathrm{fp}}(j)\,\|Z_1 - Z_2\|_{L^\infty(D)}\,\sqrt{|U|}}. \tag{11}$$

for the discrete tightness ratio against the certified per-scale bound of proposition 3.4 (here U is an interval, so $|U| = \mathrm{diam}(U)$ by remark 2.1). Then ρ_w has median 5.63×10^{-3} and maximum 0.151 (fig. 2), still strictly below 1. Factoring $\rho_w = \rho_{\mathrm{op}}\rho_{\mathrm{top},w}$ with

$$\rho_{\mathrm{op}} := \frac{\|P_j^\Psi \Delta\|_{L^\infty(D)}}{C_{\mathrm{fp}}(j)\|\Delta\|_{L^\infty(D)}}$$

and

$$\rho_{\mathrm{top},w} := \frac{\|\Delta\lambda\|_{2,w}}{\|P_j^\Psi \Delta\|_{L^\infty(D)}\sqrt{|U|}}$$

gives medians 0.091 and 0.084 and maxima 0.448 and 0.535, respectively. At the median the two factors are comparable (0.091 versus 0.084): generic slack is shared between the smoother and the landscape step, not concentrated in either. At the maxima the landscape factor retains more (0.535 versus 0.448). Every one of the 1280 comparisons has $\lambda_1 \not\equiv 0$ on $U = (-1, 1)$. The cubical complex for these 1280 comparisons is GUDHI's `CubicalComplex`, constructed from the top-dimensional-cell array using GUDHI's documented filtration convention. The near-saturating example below uses a separate 8-neighbor pixel union-find implementation. We treat these as distinct computational complexes and do not compare their persistence intervals. Every reported input pair is evaluated using one fixed complex and filtration convention.

A deterministic sign-pattern example was designed to satisfy the saturation criterion of proposition 3.3 on a 48×48 periodic grid at $j = 0$ (module `tws_validation.sign_pattern_extremal`, deterministic, no random seed). Sublevel H_0 was computed by an 8-connectivity union-find implementation using finite bars only. In floating-point arithmetic, the operator factor and the full ratio differed from one by less than 10^{-13} (recorded in the repository CSV as `rho_op`; the operator factor `op_saturation` is likewise 1). The same construction on larger grids remains at 1 through $j = 4$ (residual $< 10^{-5}$ at $j = 3$ on 128×128 and $< 10^{-6}$ at $j = 4$ on 256×256). This is strong numerical evidence that the specified discrete example saturates the bound, but it is not an exact or machine-checked proof that the persistence pairing and death value remain unchanged. By contrast, an amplitude-shift pair on a single Gaussian bump, moving the death of an H_1 tent rather than the birth (`tws_validation.extremal_tightness`, same grid and j), reaches only $\rho_{\mathrm{op}} = 0.510$ and is not an extremizer. The archival deposit will include the arrays and all intermediate persistence pairs needed to inspect this numerical near-saturation claim.

Remark 6.1 (Redundancy, not idempotence). The implemented map satisfies $P_j^{\Psi} \circ P_j^{\Psi} \neq P_j^{\Psi}$ in general, as expected of an undecimated low-pass approximation built from a redundant transform. No statement in this paper uses idempotence.

Remark 6.2 (Scope of the empirical checks). The synthetic experiments test the implemented kernels, persistence conventions, and numerical evaluation of the bound. They illustrate that generic inputs exhibit substantial slack and that a deliberately constructed input can approach saturation. Because the landscape integrals and persistence computations are numerical, we refer to these results as reproducible checks rather than formal certificates or validations of the theorem.

6.2 Numerical illustration on chest radiographs with artificial bounded perturbations

To illustrate the magnitude of Theorem A on real image arrays, we apply the pipeline to a deterministic subset of 40 public chest radiographs from the test split of Kermany et al. [22]. The experiment is not a clinical validation and does not estimate the physical noise distribution of the imaging system. Instead, we add artificial independent pixel perturbations with a prescribed L^{∞} bound.

The subset contains 20 images labeled NORMAL and 20 labeled PNEUMONIA and was selected by a pseudorandom procedure with fixed seed 20260715; the exact filename manifest is provided in `paper/medical_application/medical_subset_manifest.json`. Each image is resized to 256×256, scaled to $[0, 1]$, and inverted by $Z \leftarrow 1 - Z$. Persistence is computed using GUDHI's `CubicalComplex` on top-dimensional cells, with finite intervals only. The first

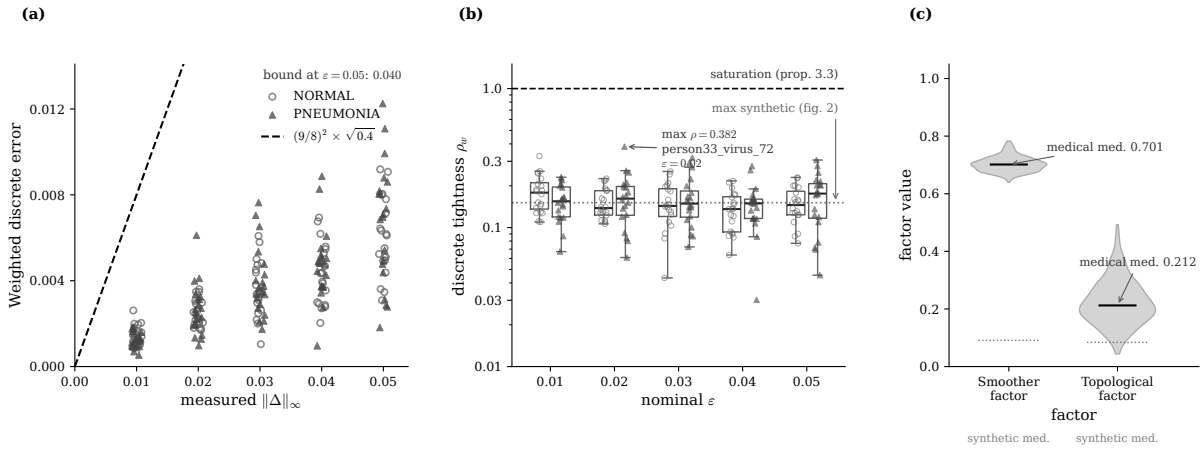


Figure 3: Numerical illustration on 40 chest-radiograph arrays with 200 artificially perturbed comparisons. Panel (a) compares the weighted discrete landscape error with the deterministic upper-bound expression; panel (b) shows the computed tightness ratios; panel (c) shows their empirical factorization. These floating-point implementation results are not clinical, statistical, or probabilistic robustness guarantees.

landscape is evaluated on a 200-node grid over $U = [0.4, 0.8]$. The preconditioner is the periodized level-one seven-tap `db2` operator from proposition 3.4.

For each nominal amplitude $\varepsilon \in \{0.01, \ldots, 0.05\}$, artificial perturbations are sampled independently from $\mathrm{Unif}(-\varepsilon, \varepsilon)$ and the perturbed array is clipped to $[0, 1]$. Because clipping changes the realized perturbation, every denominator uses its measured norm $\|\Delta\|_\infty$ rather than the nominal value ε. The perturbation generator, image ordering, and seed-assignment rule are recorded in the repository so that every image–ε comparison is deterministically reproducible.

All 200 values of the weighted-discrete ratio ρ_w of eq. (11) were below one. Their median was 0.149, their 90th percentile was 0.226, and their maximum was 0.382. These values describe this fixed subset, perturbation generator, preprocessing pipeline, and numerical quadrature only. They do not estimate a population-level robustness probability and do not compare diagnostic performance between the two image labels. Per-class medians and maxima are in table 4.

At nominal $\varepsilon = 0.05$, the worst-case expression obtained by replacing the measured perturbation norm by ε is $(9/8)^2 \times 0.05 \times \sqrt{0.4} \approx 0.0400$. The actual certificate for each row uses the measured $\|\Delta\|_\infty$ and may therefore be slightly smaller. The factorization $\rho_w = \rho_{\mathrm{op}} \rho_{\mathrm{top},w}$ has empirical medians 0.701 and 0.212, respectively. In this experiment, most of the observed slack occurs in the persistence-landscape step. We do not interpret this factorization as evidence about physical sensor noise, pathology, or clinical robustness.

7 Discussion

Theorem A bounds the composed map $Z \mapsto \lambda_n|_U \circ P_j^\Psi Z$ in $L^2(U)$. Predecessors refine the metric on diagrams or landscapes and do not see P_j^Ψ or restriction to U (table 2). The subsections that follow record boundary caveats, open problems, and a deterministic-transfer reading.

Table 4: Medical illustration of Theorem A: median and maximum of the weighted discrete ratio ρ_w of eq. (11) by class and ε (all entries to three decimals).

class / ε	0.01		0.02		0.03		0.04		0.05	
	median	max	median	max	median	max	median	max	median	max
NORMAL	0.178	0.327	0.138	0.225	0.143	0.253	0.136	0.217	0.145	0.230
PNEUMONIA	0.155	0.230	0.162	0.382	0.149	0.318	0.149	0.277	0.176	0.306
all	0.160	0.327	0.147	0.382	0.145	0.318	0.138	0.277	0.156	0.306

7.1 Boundary handling

Boundary conditions affect P_j^Ψ near ∂D through the support of the kernel. Empirically, the three folding modes of section 6.1 share the same maximal $C_{\mathrm{fp}}(0)$ at $j = 0$, the exact operator norm of eq. (9), attained on an interior row. The non-periodic modes are emulated by padding and then applying the periodic transform, with a pad narrower than the `db2` kernel support at coarse scales, so a residual periodic contribution is not excluded. A systematic boundary study, on natural images and with padding wide enough to decouple the modes, remains future work.

7.2 Open problems

1. Analogues of Theorem A for persistence images [1] and silhouettes replace $L_\lambda = 1$ by the relevant Lipschitz constant.
2. Application-domain checks on natural images and volumes, with a noise model that is not bounded in L^∞ almost surely; the artificial bounded-perturbation radiograph illustration of section 6.2 is a first step in this direction.

7.3 Deterministic transfer under bounded perturbations

Suppose $Z_1, \ldots, Z_N$ are observations of a reference field $Z^\star$ satisfying $\|Z_i - Z^\star\|_{L^\infty(D)} \le \varepsilon$ almost surely. For $\Phi_i = \lambda_n^U(P_j^\Psi Z_i)$ and $\Phi^\star = \lambda_n^U(P_j^\Psi Z^\star)$, Theorem A gives the deterministic bound

$$\|\Phi_i - \Phi^\star\|_{L^2(U)} \le C_{\mathrm{fp}}(j)\, \varepsilon \sqrt{|U|}$$

almost surely for every i.

This statement transfers an assumed input-space perturbation radius to an output-space radius. It is not a concentration inequality, does not quantify the bias $\|\mathbb{E}[\Phi_i] - \Phi^\star\|$, and does not determine how large N must be. Deriving sample-complexity bounds would additionally require a specified estimator, a probability model, and a concentration or empirical-process argument. We leave that statistical development to future work.

For illustration only, in dimension $d = 2$ at $j = 0$, with $\varepsilon = 0.01$ and $|U| = 2$, the deterministic radius is $(9/8)^2 \times 0.01 \times \sqrt{2} \approx 0.0179$. This number quantifies sensitivity to the assumed L^∞ perturbation bound; it should not be interpreted as a required sample size or as a confidence radius without additional statistical assumptions.

8 Conclusion

We gave an explicit stability estimate for a pipeline that applies an undecimated low-pass `db2` operator before computing an indexed persistence landscape on a bounded parameter interval. The estimate follows by composing standard bottleneck and landscape stability with an L^∞ operator bound and the elementary conversion from $L^\infty(U)$ to $L^2(U)$.

For the specified level-one periodic `db2` implementation on grids of side at least seven, the approximation kernel has the exact one-dimensional ℓ^1 norm $9/8$, yielding the exact d-dimensional L^∞ operator norm $(9/8)^d$. The Lean development verifies this level-one algebraic calculation and the propagation of explicitly stated abstract stability hypotheses. It does not formalize bottleneck stability, persistence landscapes, or the complete numerical pipeline.

Reproducible synthetic experiments illustrate both generic slack and near-saturation under a deliberately constructed input. A separate chest-radiograph experiment illustrates the numerical size of the bound under artificial bounded pixel perturbations; it is not a clinical validation or a physical sensor-noise model.

The main outcome is therefore a transparent, implementation-specific deterministic sensitivity bound. Extensions to other landscape summaries, boundary operators, unbounded stochastic noise, and statistically meaningful confidence or sample-complexity results require additional analysis.

Data and code availability

At the time of this manuscript version, the source tree and formalization have not yet been deposited in a public archival repository. Consequently, the numerical and Lean claims are not independently reproducible from this document alone. Before submission, we will deposit the exact source revision, environment lockfiles, generated CSV/JSON files, deterministic input arrays, and Lean build instructions in a public repository and an archival service, and we will replace this paragraph with the permanent URL, DOI, and commit hash.

The chest radiographs are not redistributed. We provide only the deterministic filename manifest and the scripts needed to process images obtained independently from the original public source, subject to its access and licensing conditions. The medical illustration of section 6.2 uses a deterministic 40-image subset (20 NORMAL, 20 PNEUMONIA) of the public test split of Kermany et al. [22]. The self-contained kit `paper/medical_application/` regenerates its figure, caption, and tables from frozen CSV/JSON via `verify_results.py`, `render_final_figure.py`, and `check_caption.py`. All other numerical inputs are synthetic. Software is MIT; the manuscript is CC BY 4.0.

Software versions. Lean 4.33.0; mathlib revision `db584cd6d46c92f209a44c0f1c829460d327499d`; Python 3.14.3; NumPy 2.5.2; PyWavelets 1.9.0; GUDHI 3.13.0. The complete environment is fixed by `experiments/requirements.lock.txt` in the archived repository.

Computational cost. For a fixed filter length, one undecimated filtering stage is linear in the number of pixels. The cost of evaluating the certificate after filtering is also linear, since it requires the measured input supremum norm and a precomputed operator bound. Persistent-homology

computation is unchanged by the certificate. We do not claim a universal runtime ordering between filtering and persistence across all implementations and input sizes.

Competing interests

The authors declare no competing interests.

A Supporting estimates and proofs

A.1 Proof of Lemma 3.1

Bubenik's theorem [2] yields $\|\lambda_n(Z) - \lambda_n(Z')\|_{L^\infty(\mathbb{R})} \le \|Z - Z'\|_{L^\infty(D)}$ after composing with the bottleneck stability of Cohen–Steiner et al. [11]. Restricting to a bounded interval U and applying eq. (1) gives

$$\|\lambda_n^U(Z) - \lambda_n^U(Z')\|_{L^2(U)} \le \sqrt{\operatorname{diam}(U)}\, \|Z - Z'\|_{L^\infty(D)}.$$

A.2 Restriction to finite persistence intervals

Under the extended bottleneck metric, the distance between a point with finite death coordinate and a point with infinite death coordinate is infinite. Hence every finite-cost matching pairs finite intervals only with finite intervals or with the diagonal, and pairs essential intervals only with essential intervals. Restricting such a matching to the finite subdiagrams cannot increase its cost. Therefore,

$$d_B(\mathrm{Dgm}_k^{\mathrm{fin}}(Z), \mathrm{Dgm}_k^{\mathrm{fin}}(Z')) \le d_B(\mathrm{Dgm}_k(Z), \mathrm{Dgm}_k(Z')) \le \|Z - Z'\|_{L^\infty(D)}.$$

The ordinary persistence-landscape non-expansiveness theorem may then be applied to the finite subdiagrams used in this paper.

A.3 Certified kernel bounds

For a periodic finite-grid implementation, the exact operator norm is the maximum absolute row sum, equivalently the ℓ^1 norm of the periodized kernel in the separable convolution setting. The non-periodized quantity $\|k_j\|_{\ell^1}^d$ is always a certified upper bound:

$$\|P_{j,N}^{\Psi}\|_{L^\infty \to L^\infty} = \|k_{j,N}\|_{\ell^1}^d \le \|k_j\|_{\ell^1}^d.$$

At level $j = 0$ and grid side $N \ge 7$, no coefficient overlap occurs and the bound is exact, giving $(9/8)^d$. At higher levels we use $\|k_j\|_{\ell^1}^d$ as a certified bound unless the finite-grid row sum has been evaluated explicitly.

A coarser estimate involving separate scaling-function support and norm bounds is unnecessary and can be substantially less informative. Accordingly, no such estimate is used in Theorem A or in the numerical ratios.

A.4 Reverse triangle inequality

Lemma A.1 (Reverse triangle inequality)**.** *For $f, g : \mathbb{R} \to \mathbb{R}$ square-integrable on U, $\big|\|f\|_{L^2(U)} - \|g\|_{L^2(U)}\big| \leq \|f - g\|_{L^2(U)}$. The bound applies to $\lambda_n^U(P_j^\Psi Z)$: on a finite cubical grid the persistence barcode has finitely many intervals, so the landscape is piecewise-linear with finitely many tents and, restricted to bounded U, square-integrable; in the continuum, q-tameness yields a locally bounded landscape, still square-integrable on bounded U.*

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